

Synchronous Assessment - Breakout Case Study

Traditional Public Health Project - Digital Classroom

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Purpose: Use a live, small-group case study to apply cancer surveillance concepts and data quality dimensions in a realistic scenario.

Breakout Activity Overview

- Mode: Breakout Rooms (Zoom or similar platform)
- Group Size: 3-5 learners per room
- Time: 10 minutes of group discussion + report-out
- Materials: This handout or a single shared slide in each room

Case Scenario: The “Rural Spike” Mystery

Your state cancer registry has detected a sudden 20% increase in late-stage lung cancer cases in "County X," a largely rural county, over the last 12 months.

During the same period, the main hospital in County X implemented a new Electronic Health Record (EHR) system and changed how it sends data to the registry.

Group Task (10 Minutes)

Work together to discuss and answer the questions below. Be ready to share one concise answer per group when we return to the main room.

1. **Analyze - Real cluster or data artifact?** Discuss whether this “spike” is more likely to be a true increase in late-stage disease, a data artifact related to the new EHR, or a mix of both. List 1-2 brief reasons for your conclusion.
2. **Diagnose - Data quality dimension at risk:** Which single data quality dimension is most likely to be incorrect in this scenario? Choose one and be prepared to justify

your choice:

- Completeness
- Accuracy
- Timeliness
- Consistency

3. **Solve - Proposed data quality check:** Propose one specific, practical check you would run to verify the data behind this apparent spike. Think about comparisons, audits, or cross-checks you could perform using existing sources (e.g., pathology, EHR, registry).

Group Deliverable

When you return to the main room, one person from your group should type a short response in the Zoom chat using this format:

- Diagnosis (Data Quality Dimension): _____
- Proposed Check (1-2 sentences): _____

Keep your answer to 2-3 sentences total.

Answer Key (For Grading)

This section is for instructor use only and should not be displayed to learners during the activity.

Likely Diagnosis

Most defensible choices:

- Accuracy - especially staging accuracy for late-stage lung cancer; and/or
- Consistency - mapping issues between the new EHR fields and registry data items.

A new EHR may change how stage, primary site, or diagnosis dates are recorded and transmitted, leading to apparent shifts that reflect coding or mapping changes rather than true changes in disease burden.

Reasonable Data Quality Checks

Examples of acceptable “Proposed Check” answers include:

- Re-abstraction audit:
 - Re-abstract a sample of late-stage lung cancer cases from County X using source documents (pathology reports, clinician notes, imaging) and compare them to the registry records.
- Pathology vs registry comparison:
 - Compare pathology reports directly to the registry’s coded stage, primary site, and diagnosis date for the same cases to identify systematic miscoding.

Other possibilities could include examining trends in neighboring counties, checking pre- vs post-EHR implementation data, or reviewing the EHR-to-registry interface specifications for stage and diagnosis fields.
